Sustech 2025南科大入学考试物理样卷-更新

NO. 1

Mark scheme

D

NO. 2

Mark scheme

D

NO. 3

Mark scheme

A

NO. 4

Mark scheme

C

NO. 5

Mark scheme

D

NO. 6

Mark scheme

C

NO. 7

Mark scheme

A

NO. 8

Mark scheme

D

NO. 9

Mark scheme

D

NO. 10

Mark scheme

B

NO. 11

Mark scheme

C

NO. 12

Mark scheme

C

NO. 13

Mark scheme

D

NO. 14

Mark scheme

A

NO. 15

Mark scheme

C

NO. 16

Mark scheme

Question	Answer	Marks
a.i	circles drawn around any two particles with seven (uncircled) particles in between	A1
a.ii	curve has an initial negative displacement and initial amplitude same as original curve	B1
	curve has same amplitude as original curve throughout	B1
	curve has same wavelength as original curve throughout, with constant (non-zero) phase difference	B1
a.iii	(to the) right / rightwards	A1
b	$\lambda = 2 \times 0.19$ $= 0.38$ m	C1
	$v = f\lambda$	C1
	$v = (16 \times 10^3) \times 0.38$ $v = 6100$ m s^{-1}	A1
c	$I \propto A^2$	C1
	$I = (1/2)^2 I_0$ intensity = 0.25 I_0	A1
d.i	same frequency / wavelength / period	B1
d.ii	• a stationary wave has nodes/antinodes (and a progressive wave does not) • a stationary wave does not transfer/propagate energy (and a progressive wave does transfer/propagate energy) • different points on a stationary wave have different amplitudes (and all points on a progressive wave have the same/constant amplitude) • stationary wave has adjacent particles that are in phase (and adjacent particles on progressive wave are out of phase)	B2

	Any two points, 1 mark each. Allow the reverse statement for each marking point.		

NO. 17

Mark scheme

a	minimum frequency (of light) for electron(s) to be emitted (from surface)	M1
	reference to frequency of electromagnetic radiation/ photon	A1
	or	
	frequency causing emission of electron(s) from surface with zero kinetic energy	(M1)
	reference to frequency of electromagnetic radiation/ photon	(A1)
b.i	positive intercept on (1/ λ)-axis (when extrapolated)	B1
	straight line with positive gradient	B1
b.ii	gradient = hc where c is the speed of light	B1
b.iii	maximum kinetic energy when electron emitted from surface	B1
	energy is required to bring an electron to the surface	B1
b.iv	each photon has more energy	M1
	fewer photons per unit time	M1
	fewer electrons per unit time/ less current	A1

NO. 18

Mark scheme

Question	Answer	Marks	
a	$E = 0$ or $E_A = (-)E_B$ (at $x = 11$ cm)	B1	
	$Q_A / x_2 = Q_B / (20 - x)^2 = 11^2 / 9^2$	C1	
	Q_A / Q_B or ratio = 1.5	A1	[3]
	or		
	$E \propto Q$ because r same *or* $E = Q/ 4\pi\varepsilon_0 r^2$ and r same	(B1)	
	$Q_A/Q_B = 48/32$	(C1)	
	Q_A / Q_B or ratio = 1.5	(A1)	
b.i	for max. speed, $\Delta V = (0.76 - 0.18)$ V or $\Delta V = 0.58$ V	C1	
	$q\Delta V = ½mv^2$ $2 \times (1.60 \times 10^{-19}) \times 0.58 = ½ \times 4 \times 1.66 \times 10^{-27} \times v^2$	C1	
	$v2 = 5.59 \times 10^7$ $v = 7.5 \times 10^3$ ms^{-1}	A1	[3]
b.ii	$\Delta V = 0.22$ V	C1	
	$2 \times (1.60 \times 10^{-19}) \times 0.22 = ½ \times 4 \times 1.66 \times 10^{-27} \times v^2$ $v^2 = 2.12 \times 10^7$ $v = 4.6 \times 10^3$ ms^{-1}	A1	[2]

NO. 19

Mark scheme

Question	Answer	Marks	
a.i	0.225 s <u>and</u> 0.525 s	A1	[1]
a.ii	period or $T = 0.30$ s <u>and</u> $\omega = 2\pi / T$	C1	
	$\omega = 2\pi/0.30$ $\omega = 21$ rad s^{-1}	A1	[2]
a.iii	speed = ωx_0 or $\omega(x_0{}^2 - x_2)^{1/2}$ <u>and</u> $x = 0$	C1	
	$= 20.9 \times 2.0 \times 10^{-2} = 0.42$ ms^{-1}	A1	[2]
	or		
	use of tangent method: correct tangent shown on Fig. 4.2	(C1)	
	working e.g. $\Delta y / \Delta x$ leading to maximum speed in range (0.38–0.46)ms^{-1}	(A1)	
b	sketch: reasonably shaped continuous oval/circle surrounding (0,0)	B1	
	curve passes through (0, 0.42) and (0, –0.42)	B1	
	curve passes through (2.0, 0) and (–2.0, 0)	B1	[3]